Francesco Sannicola — Machine Learning Engineer | Data Scientist

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Work Experience

Data Scientist II — ***Agile Lab*** — Sep 2025 – Present

- Developed embedding-based **customer-value models** to identify cross-selling opportunities and predict savings propensity.
- Engineered a geospatial routing engine on a multi-layer graph for infrastructure planning, with dynamically generated filters, supported by spatial indexing and multi-level **caching**—reducing worst-case query time from 2 minutes to 30 seconds.

Machine Learning Engineer — Senior Consultant — ***Links M&T*** — Jan 2025 – Aug 2025

- Built multi-tenant IoT energy forecasting platform for 3 clients. Used genetic algorithms, **MLOps** (tracking, registry, CI/CD), and real-time dashboards. Cut peak costs, consumption, and CO_2 via per-tenant demand forecasts.

Machine Learning Engineer — Consultant — ***Links M&T*** — Mar 2023 – Dec 2024

- Integrated browsing, crawling, GraphRAG into **ReAct agent chatbot**. Unified on-premise/online sources; cut reporting prep by 15–30 min/session.
- Designed ESG reporting with **RAG** + **GPT**. Automated KPI extraction from fund reports; lowered compliance costs.
- Developed an **NLP** web application powered by **encoder-only transformers** for contextual content recommendations with sub-5s latency, including speech-to-text features.
- Engineered **predictive maintenance** for EV chargers. Degradation models cut costs 30%, prevented failures.
- Developed loan **default prediction** model (80% accuracy, $<$2% false positives). Flagged risks 3–6 months early.
- **Lecturer** in software development, databases, and Python at *Associazione Formazione*, and in anomaly detection and time series analysis at *Links Academy*. Resulted in 7+ student job placements; practical lab materials available on *GitHub*.

Artificial Intelligence and Data — Analyst — ***Deloitte*** — Oct 2022 – Feb 2023

- Developed **ETL** workflow with auto KPI visualizations from surveys. Turned static reports into real-time dashboards for engagement/retention tracking.

AI Engineer and System Administrator — Intern — ***Intesa Sanpaolo*** — Dec 2021 – May 2022

- Implemented real-time **time series forecasting** + anomaly detection. Advanced from stats to RNNs/autoencoders; cut failure response from minutes to seconds.

Education

Master's Degree in Computer Science — AI GPA: 3.71/4 Jan 2020 – Apr 2022
University of Turin

- Coursework: Neural Networks; Machine Learning; Natural Language Technologies; Advanced Software Architectures.
- 200h collabs: web pages development for freshman orientation; study support for a student with disability.
- Thesis: real-time **anomaly detection** using stats, ML, DL (*thesis link*).

Bachelor's Degree in Computer Science GPA: 4/4 Oct 2016 – Dec 2019
University of Bari

- Coursework: Computer Architecture; Operating Systems; Algorithms; Databases; Software Engineering.
- Thesis: full-stack indoor positioning system integrating a content-based **recommender system**.

Skills

Programming: Python, Java, C, SQL, Bash
ML/AI: GitHub Copilot, TensorFlow, Keras, PyTorch, scikit-learn, RAG, ReAct agents, LangChain, LangGraph, Transformers
Data & Analytics: Pandas, MS Excel, Qlik Sense, Qlik NPrinting
Databases: PostgreSQL, MongoDB, Redis, Milvus
Cloud & DevOps: AWS (EC2, S3, Lambda, SageMaker, Transcribe, Polly, SQS, SNS, RDS, EventBridge), Databricks, GCP, Docker, Kubernetes, Git, GitLab CI/CD
Frameworks & Tools: FastAPI, Spring Boot, Pytest, Flask, Keycloak, Uv, Ruff, Pre-commit, Bokeh

Certifications

AWS Certified Machine Learning Apr 2024
Credentials: *Credly profile*
Courses: AWS ML Certification Preparation (Mar 2024), GCP Big Data and ML Fundamentals (May 2022), GCP Fundamentals: Core Infrastructure (Apr 2022)

Languages

Italian: Native | English: Professional Working Proficiency